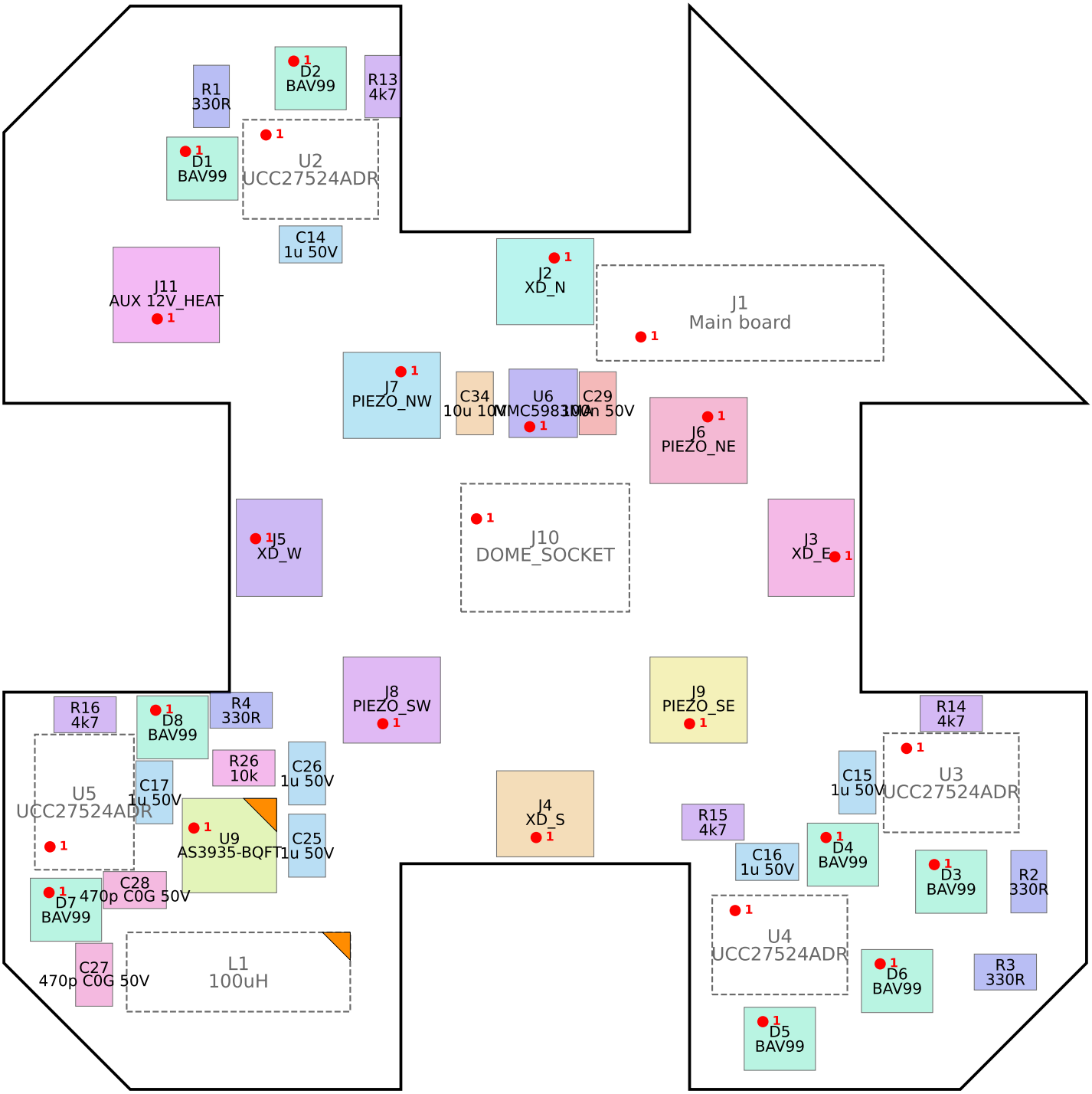
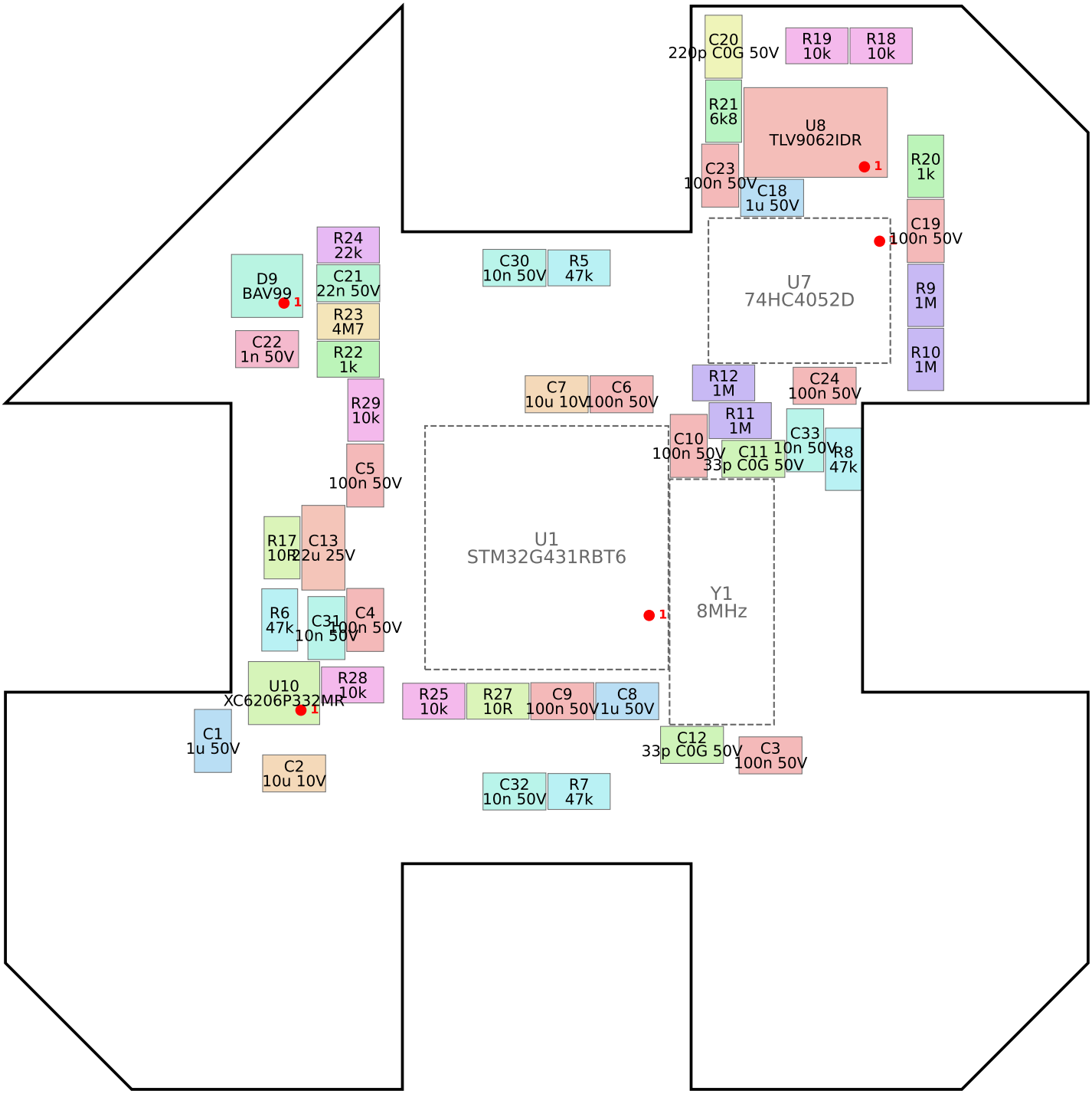
# crown - TOP

red dot 1 = pin 1  black bar = cathode  red + = cap +  orange corner = hand-fitted  dashed = fit last  (key: last page)



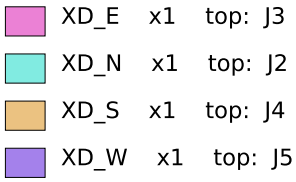
# crown - BOTTOM (seen from below, mirrored)

red dot 1 = pin 1  black bar = cathode  red + = cap +  orange corner = hand-fitted  dashed = fit last  (key: last page)



## crown - values

- 100n 50V x1 top: C29
- 100n 50V x9 bottom: C3, C4, C5, C6, C9, C10, C19, C23, C24
- 100uH x1 top: L1
- 10R x2 bottom: R17, R27
- 10k x1 top: R26
- 10k x5 bottom: R18, R19, R25, R28, R29
- 10n 50V x4 bottom: C30, C31, C32, C33
- 10u 10V x1 top: C34
- 10u 10V x2 bottom: C2, C7
- 1M x4 bottom: R9, R10, R11, R12
- 1k x2 bottom: R20, R22
- 1n 50V x1 bottom: C22
- 1u 50V x6 top: C14, C15, C16, C17, C25, C26
- 1u 50V x3 bottom: C1, C8, C18
- 220p C0G 50V x1 bottom: C20
- 22k x1 bottom: R24
- 22n 50V x1 bottom: C21
- 22u 25V x1 bottom: C13
- 330R x4 top: R1, R2, R3, R4
- 33p C0G 50V x2 bottom: C11, C12
- 470p C0G 50V x2 top: C27, C28
- 47k x4 bottom: R5, R6, R7, R8
- 4M7 x1 bottom: R23
- 4k7 x4 top: R13, R14, R15, R16
- 6k8 x1 bottom: R21
- 74HC4052D x1 bottom: U7
- 8MHz x1 bottom: Y1
- AS3935-BQFT x1 top: U9
- AUX 12V_HEAT x1 top: J11
- BAV99 x8 top: D1, D2, D3, D4, D5, D6, D7, D8
- BAV99 x1 bottom: D9
- DOME_SOCKET x1 top: J10
- MMC5983MA x1 top: U6
- Main board x1 top: J1
- PIEZO_NE x1 top: J6
- PIEZO_NW x1 top: J7
- PIEZO_SE x1 top: J9
- PIEZO_SW x1 top: J8
- STM32G431RBT6 x1 bottom: U1
- TLV9062IDR x1 bottom: U8
- UCC27524ADR x4 top: U2, U3, U4, U5
- XC6206P332MR x1 bottom: U10


XD_E x1 top: J3
XD_N x1 top: J2
XD_S x1 top: J4
XD_W x1 top: J5

# crown - key

## Marks on the board pages

red dot + 1 : pin 1 of ICs, transistors, LEDs, optocouplers, connectors and modules

black bar   : diode cathode, matching the band printed on the diode body

red +       : + pad of electrolytic / tantalum capacitors

no mark     : resistors, ceramic capacitors, inductors, fuses, crystals, pin strips, bidirectional TVS

orange corner: part fitted by hand (not placed by JLCPCB); dashed outline: large part, fit it last

the BOTTOM page is mirrored: it shows the board as you see it from below

## Finding pin 1 on the real part

SOT-23 (3 legs): the single leg on its own side is pin 3; pin 1 is left of it on the two-leg side

SOT-23-5 / SC-70 / SOIC / VSSOP / QFN / LGA: a dot, dimple or bevelled edge marks pin 1

Addressable LEDs: a marked (cut / green) corner - check the datasheet, it is easy to misread

Diodes: the band is the cathode (K)

Electrolytic capacitors: the stripe marks the - side (the other pad is +)

## Tools

0805 / SOT-23: a 3-5x magnifier or a 10x loupe

QFN / LGA / DFN / SC-70: a USB microscope (check the pads under the part after reflow)